

Vision Document

Project Title: University Micro-Internship Matching Portal

Team 2 Occ 1

Problem Statement

Overview

Universities today face challenges in connecting students with suitable internship and micro-internship opportunities. The current system is fragmented, involving multiple stakeholders such as students, lecturers, career centre staff, and company HR representatives. Instead of having one centralized platform, the process relies on a mix of external job portals (e.g., LinkedIn, JobStreet), university systems, emails, and informal communication methods.

Because of this, the overall internship process becomes inefficient, unclear, and difficult to manage for everyone involved.

Core Challenges

1. Difficulties in Opportunity Discovery (Students)

- Students often struggle to find relevant internship or micro-internship opportunities.
 - Information is scattered across different platforms, and there are limited tools to filter or match roles based on their skills and interests.
- For micro-internships in particular, short-term or project-based roles are often mixed with long-term internships, making them harder to find.
- In addition, students need to repeatedly enter the same personal and academic information across multiple systems, which is time-consuming.
- Students also face issues such as:
 - Delayed responses from companies
 - Lack of transparency in application status
 - Insufficient details about job scope or company credibility

2. Lack of Monitoring and Coordination (Lecturers & Career Centres)

- Lecturers and career centres face difficulties in managing and tracking student internships.
- There is no single system to monitor applications, placements, and student progress in real time.
- This makes it hard to:
 - Ensure students meet academic requirements
 - Verify the quality and credibility of internship providers
 - Maintain clear communication with students and employers

- In addition, tasks such as generating reports, analyzing outcomes, and maintaining industry partnerships require a lot of manual effort due to the lack of a centralized database.

3. Inefficient Candidate Matching (Employers / HR)

- Employers also face challenges in finding suitable candidates. Without a proper matching system, they often receive many applications that do not meet their requirements.
- This leads to:
 - Time spent filtering irrelevant applications
 - Difficulty identifying students with specific skills needed for short-term projects
 - Mismatch between student abilities and job expectations

4. Process Delays and Lack of Integration

- The current process involves multiple steps across different systems, which slows things down.
- For micro-internships, this is especially problematic because the approval or coordination time can be longer than the duration of the task itself.
- Overall, there is no “single source of truth” where all stakeholders can access accurate and updated information.

Conclusion

The current internship ecosystem lacks efficiency, transparency, and coordination. These issues affect students, lecturers, career centres, and employers alike.

Therefore, there is a need for a centralized and user-friendly platform that:

- Brings all internship information into one place
- Simplifies the application process
- Improves matching between students and opportunities
- Supports better communication and tracking for all stakeholders

This project aims to develop a **University Micro-Internship Matching Portal** to address these challenges and improve the overall internship experience.

Vision Statement

To create a smart and accessible internship platform that connects students with the right opportunities, enabling career growth while helping HR professionals discover emerging talent efficiently

Stakeholder and User Personas

Stakeholder Role	Org. Position	Key Responsibilities	User Type	Primary Goal
Project Facilitator	Senior Career Development Officer	Vet companies and ensure university compliance.	Super Admin	Scale placements without increasing manual workload.
Industry Partner	HR Executive	Post micro-internship positions and monitor intern performance.	Employer Admin	Find and hire qualified talent quickly.
Academic Monitor	Senior Lecturer	Validate technical relevance and academic credit.	Moderator	Align external projects with Course Learning Outcomes.
Industry-exposed Student	Final Year Student	Complete advanced tasks and manage professional profile.	Standard User	Secure high-level tasks to maintain industry edge.
Pre-internship Student	Second Year Student	Apply for beginner tasks and build initial portfolio.	Standard User	Gain confidence through "safe-to-fail" work.

Senior Career Development Officer

Puan Siti



Age 43

Occupation Senior Career Development Officer

Year of Experience 15 years in higher education administration and industry relations

Tech Literacy

Goal

- Increase student completions without adding manual overhead.
- Ensure micro-internships deliver genuine educational value.
- Generate instant employment metrics for university management.

Pain Points

- Student and company records sit in separate, disconnected silos.
- Confirmation of project completion requires constant email follow-ups.
- Verifying company legitimacy is a manual, time-consuming process.

Key Responsibilities

- Maintaining a pool of reliable company partners.
- Ensuring micro-internships meet university guidelines.
- Support students and resolve issues with their industry mentors.

Motivations

- Seeing students gain confidence through short, manageable tasks.
- Reducing the "back-and-forth" emails between the university and HR representatives.

HR Executive

Puan Aisyah



Age 34

Occupation Human Resources Executive

Year of Experience 8 years in recruitment and internship coordination

Tech Literacy

Goal

- Find suitable university students for short-term micro-internship opportunities quickly
- Reduce time spent filtering unsuitable applicants
- Ensure students selected match company skill requirements

Pain Points

- Receiving too many applications that do not meet company requirements
- Difficulty tracking student progress during internship
- Inconsistent student profiles and incomplete application details
- Delays in approval and scheduling

Key Responsibilities

- Post available micro-internship positions
- Review student applications
- Shortlist candidates based on skills and academic fit
- Coordinate interview schedules
- Approve internship placements
- Monitor intern attendance and performance

Motivations

- Wants efficient recruitment with less administrative burden
- Seeks high-quality candidates who can contribute immediately
- Wants easy access to student skill portfolios and academic background

Senior Lecturer

Dr. Aris



Age 43

Occupation Senior Lecturer & Industry Engagement Coordinator

Year of Experience 10 years in academia, with a background in Agile Project Management

Tech Literacy

Goal

- Ensure that the micro-internships students take are technically relevant to the current curriculum.
- Use real-world project outcomes as case studies or practical assessments for his Software Architecture and Design Paradigm course.
- Reduce the time spent manually validating if a student's external work qualifies for academic credit or soft skill points.

Pain Points

- Students often perform tasks that are too easy or unrelated to their major, making it hard to justify as learning.
- No central view of what his advisees are doing; he only finds out they did a micro-internship when they ask for a signature at the end of the semester.
- Difficulty communicating with industry mentors to get a technical critique of the student's work.

Key Responsibilities

- Moderating the technical quality of internship offers.
- Mentoring final-year students on their career paths.
- Verifying that micro-internship completion aligns with university-mandated Learning Outcomes (CLOs).

Motivations

- Seeing theory meet practice when a student comes to class and says, "We actually used this design pattern in my micro-internship last week."
- Improving the faculty's Graduate Employability (GE) rating through early industry exposure.
- Building his own professional network for potential research collaborations with participating companies.

Final Year Student

Julian



Age 22

Occupation Final Year Software Engineering Student

Year of Experience 6 months (Completed compulsory industrial training)

Tech Literacy

Goal

- Maintain industry connections and keep his technical skills "warm" while finishing his final year thesis.
- Earn extra income through high-level micro-tasks that utilize the specific skills (e.g., React, Cloud Ops) he learned during his 6-month internship.
- Stack specialized micro-credentials to negotiate a higher starting salary for his upcoming full-time job search.

Pain Points

- Most university career platforms only show entry-level or basic roles that feel like a step backward after his 6-month professional experience.
- He has very limited deep work blocks available because of his Final Year Project (FYP) and senior-level seminars.
- He finds it frustrating to re-upload his 6-month internship certificate and CV every time he wants a small task.

Key Responsibilities

- Completing a high-quality Final Year Project (FYP).
- Mentoring junior students or participating in student tech leads.
- Actively scouting for Graduate Program openings while using micro-internships to bridge the gap.

Motivations

- Wants projects where he can work independently without heavy supervision.
- Worried that if he only focuses on theory for his final year, he'll lose the edge he gained in the industry.
- Wants to leverage his existing professional reputation to skip the "basic" vetting process.

Second Year Student

Maya



📅 Age

21

🎓 Occupation

Second Year Software Engineering Student

🕒 Year of Experience

0 years (No professional experience yet)

🔧 Tech Literacy

🎯 Goal

- Build a portfolio that demonstrates he can handle real-world tasks, not just classroom assignments.
- Boost his confidence by successfully completing a task for a real company.
- Earn a small stipend to help with daily university expenses.

💡 Pain Points

- He feels intimidated by Job Descriptions that list 10+ required tools she hasn't fully mastered yet.
- He isn't sure which career path (Frontend, Backend, UI/UX) suits her best.
- He fears rejection because she has no "proof" of her technical competency.

📋 Key Responsibilities

- Maintaining a high CGPA in foundational core subjects.
- Balancing club activities (e.g., Google Developer Student Club) with academic workload.
- Self-studying frameworks (like Flutter or React) via YouTube and Coursera.

💪 Motivations

- He wants projects where he can learn while doing, ideally with a mentor who provides constructive feedback.
- Getting a "verified" badge for a basic skill (like HTML/CSS or Bug Testing) makes him feel industry-ready.
- A 5-hour task feels much safer than a 3-month commitment for a first-timer.

System Overview

The **University Micro-Internship Matching Portal** is a centralized, digital ecosystem designed to bridge the gap between academic learning and industry application. By consolidating fragmented processes into a single source of truth, the portal streamlines how students discover project-based work, how lecturers oversee academic relevance, and how employers source emerging talent.

The Integrated Workflow

The portal functions as a tripartite engine, synchronizing the needs of three distinct user groups:

- **For Students (The Talent Hub):** It acts as a smart discovery engine. Instead of repetitive data entry across multiple sites, students maintain a single professional profile. The system uses skill-tagging and filters to highlight short-term tasks that match their specific academic discipline, providing real-time transparency into application statuses.
- **For Lecturers & Career Centres (The Governance Layer):** It serves as a monitoring dashboard. Lecturers act as Moderators who can validate that a project aligns with Course Learning Outcomes (CLOs). The system automates the verification of company credibility and generates real-time reports on student progress, replacing manual email chains with a unified tracking interface.

- **For Employers (The Talent Sourcing Tool):** It functions as a precision hiring tool. HR representatives can post niche, project-based roles and utilize candidate matching logic to filter for specific technical skills. This reduces the time wasted on irrelevant applications and provides direct access to a verified pool of university talent.

Capability	Description
Centralized Repository	A single database for all job postings, student profiles, and academic records.
Smart Filtering	Advanced logic that separates long-term internships from micro-projects based on duration and skill tags.
Real-Time Monitoring	A live dashboard for lecturers to track placements and ensure university compliance.
Integrated Communication	In-platform notifications and status updates between HR, students, and supervisors.